Multi-Level Product Classification in E-Commerce Using Machine Learning: A Case Study on Etsy

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***Abstract*—This project focuses on predicting product categories for Etsy, a global marketplace for unique and creative goods. The task involves classifying products into top-level and bottom-level categories based on their metadata, such as titles, descriptions, tags, and attributes like material, color, and occasion. Using machine learning techniques, we analyze patterns in the training dataset to build models capable of accurately predicting these categories for unseen products. Additionally, we explore visualizing learned representations to demonstrate how similar items cluster together and compare the effectiveness of pre-trained embeddings with fine-tuned models. The goal is to maximize the F1 score for both top-level and bottom-level category predictions, contributing to improved organization and searchability on e-commerce platforms like Etsy.**

***Index Terms*—product categorization, e-commerce, text classification, embeddings, pre-trained embeddings**

I. Introduction

E-commerce platforms have transformed the way people shop, offering a vast array of products tailored to individual preferences. The rapid growth of online shopping has created new opportunities and challenges in managing vast product catalogs through intelligent automation. Among these platforms, Etsy stands out as a global marketplace for unique and creative goods, connecting nearly 100 million buyers with 7.7 million sellers worldwide. Etsy's focus on handcrafted, vintage, and personalized items makes it a rich environment for machine learning challenges, particularly in product categorization.

Product categorization is a critical component of modern e-commerce systems, enabling improved search relevance, personalized recommendations, and efficient inventory management. As online marketplaces like Etsy continue to scale, manual categorization becomes infeasible due to the sheer volume and diversity of listings. This challenge is further compounded by the noisy and often ambiguous metadata provided by users, such as titles, descriptions, and tags. Automating this task requires models that can understand both semantic content and contextual cues while handling high-dimensional and sparse data.

The goal of this project is to predict top-level and bottom-level category labels for Etsy product listings using a dataset containing metadata such as titles, descriptions, tags, materials, colors, and other attributes. To tackle this multi-class classification problem, we employ a combination of traditional machine learning techniques and advanced deep learning methods, including pre-trained embeddings (e.g., Sentence-BERT) and TF-IDF vectorization. These features are integrated into models such as XGBoost and neural networks to achieve state-of-the-art performance in predicting both coarse-grained and fine-grained categories.

Additionally, we explore visualizing learned representations using t-SNE to demonstrate clustering of similar items and compare the performance of pre-trained embeddings with fine-tuned models. The ultimate objective is to maximize the F1 score for both top-level and bottom-level category predictions, contributing to improved organization and searchability on large-scale e-commerce platforms like Etsy.

Related Work

Bi et al. [1] proposed a multimodal late fusion model combining textual and visual data to enhance product classification, showing improved accuracy over traditional single-modality models. Similarly, Mohamed and Hamdy [2] evaluated transformer-based architectures like BERT, demonstrating their superior performance compared to classical approaches. These works underscore the shift toward deep learning and multimodal strategies in e-commerce classification, where capturing both semantic and visual cues leads to richer representations.

Building on transformer models, Ozyegen et al. [3] introduced a dynamic masking mechanism for hierarchical categorization, addressing the complexity of multi-level taxonomies. In parallel, Kumar et al. [4] leveraged transfer learning with pre-trained BERT models, achieving better accuracy and reduced training time, which is crucial in data-scarce real-world environments. Together, these studies emphasize how transformer models can be fine-tuned and customized to reflect domain-specific structures and support scalability across varying inventory depths.

While transformer-based solutions dominate recent research, classical machine learning techniques remain relevant, particularly for resource-constrained scenarios. Jayadharshini et al. [5] applied lightweight models with NLP-based features for medium-scale platforms, offering a balance between performance and computational efficiency. Kamath et al. [6] complemented this by evaluating embedding strategies like Word2Vec, GloVe, and FastText, highlighting the importance

of semantic feature quality in boosting classification outcomes. These findings support the view that proper feature representation, even with simpler models, can yield competitive results when fine-tuned for the domain.

Zhang et al. [7] presented a broad survey of ML and DL techniques across e-commerce applications, identifying challenges and trends in classification, search, and recommendation. Their work provides a valuable map of the field's evolution. Cotacallapa et al. [8] addressed taxonomy structure through a flat-hierarchical model, bridging the gap between flat and fully hierarchical classifiers to enhance scalability and interpretability. These structural approaches are particularly useful for maintaining consistent classification in large, evolving catalogs.

In terms of pipeline optimization, Reddy et al. [9] focused on the role of preprocessing and feature engineering, revealing their impact on noisy, real-world product data. Complementing this, Motwani et al. [10] introduced a cloud-native, real-time classification framework suitable for large-scale deployments, emphasizing operational scalability in dynamic e-commerce settings. The emphasis on system architecture and deployment pipelines reflects a growing interest in making ML models production-ready.

Although some studies focus outside e-commerce, they offer transferable insights. Kim and Gil [11] used TF-IDF and LDA for document classification, methods that remain effective where deep learning isn't feasible or interpretable models are preferred. Fortuna et al. [12] explored dataset bias and annotation challenges in hate speech detection, issues also pertinent to e-commerce datasets with user-generated content and subjective labels. These lessons inform robust data collection and annotation practices in product classification tasks.

Hybrid and cross-domain approaches are gaining traction in dealing with diverse data types and complex input structures. Alagarsamy and James [13] proposed a dual-layer model combining LSTM and XGBoost for complex text, demonstrating effectiveness on structured data with embedded semantics. Fortuna et al. [14] explored generalization in toxicity detection, advocating for diverse training data a key consideration in multilingual and multicultural marketplaces. These approaches push for models that not only perform well in controlled environments but also generalize across product domains and user populations.

Finally, Cheng et al. [15] introduced a dual-expert framework integrating semantic and taxonomy-aware classifiers, setting a new benchmark for adaptable product classification. Their work reinforces the trend toward intelligent, modular systems capable of handling nuanced and large-scale inventories in e-commerce. As product metadata grows increasingly complex and heterogeneous, such modular architectures offer the flexibility to handle domain shifts, hierarchical constraints, and multi-source features in a unified framework.

Overall, the existing literature highlights a steady evolution from classical models to deep learning and hybrid approaches, emphasizing both performance and scalability.

II. Methodology

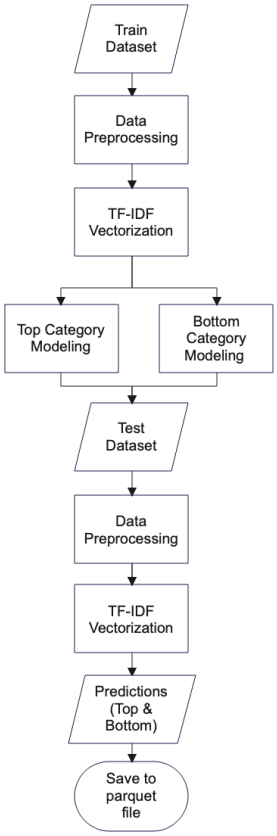

Fig. 1. **Workflow for Top and Bottom Category Prediction**

A. *Data Preprocessing*

The dataset contains 228,061 rows and 23 columns, including textual, categorical, and numerical features. Initial analysis revealed significant missing values in several columns, particularly in categorical features like `room`, `craft_type`, and `recipient`. Below are the key preprocessing steps:

*1) **Handling Missing Values**:* Missing values were handled as follows: - For textual features (`title`, `description`, `tags`), missing values were imputed with placeholders such as '"No Tags"'. - Highly missing categorical features (e.g., `shape`, `art_subject`) were dropped to avoid noise and sparsity issues. - Remaining categorical features were encoded using Label Encoding.

*2) **Feature Selection**:* Columns with extremely high proportions of missing values were removed to improve model stability. The final dataset retained key features, including textual data, categorical attributes, and derived numerical features such as title length, description length, and tag count.

*3) **Textual Feature Analysis**:* The most frequent tags included terms like `vintage`, `handmade`, and `gift`. These tags provide valuable insights into product characteristics and were leveraged for category prediction.

*4) **Numerical Feature Analysis**:* Summary statistics of tag lengths revealed: - Mean: 10.47 - Standard Deviation: 4.32 - Median: 13 - Maximum: 13

These statistics were used to derive additional features, such as normalized tag length, which helped capture variations in user-generated content.

*5) **Class Imbalance**:* The distribution of top-level categories showed significant imbalance, with `home_and_living` being the most frequent category (over 50,000 samples) and `pet_supplies` being the least frequent (around 5,000 samples).

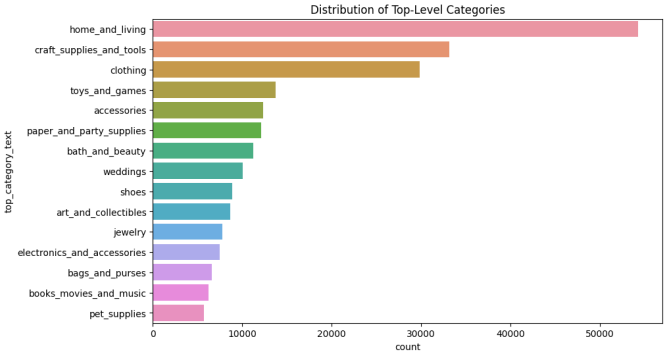

Fig. 2. **Top-Level Product Category Distribution**

*6) **Correlation Analysis**:* A correlation matrix was computed to analyze relationships between numerical features and the target variable. Stronger correlations were observed between text length features, indicating their potential predictive power.

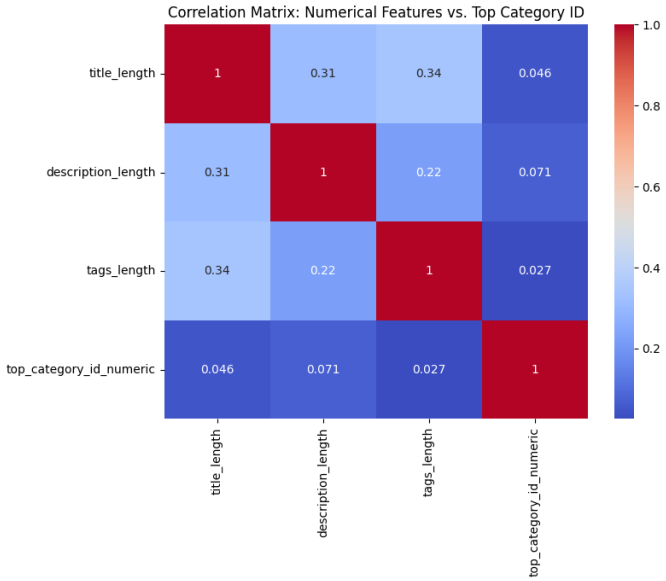

Fig. 3. **Correlation Heatmap of Numerical Feature Lengths with Top Category ID**

B. *Modeling*

The modeling process starts with the training dataset, which first goes through a cleaning and preprocessing phase. This involves standard text processing steps like removing punctuation, converting everything to lowercase, and filtering out common stopwords to make the data more consistent. Once the text is cleaned, it's transformed into numerical features using TF-IDF vectorization a method that highlights important words by considering how often they appear in a product listing compared to how common they are across all listings.

With the data ready, two separate machine learning models are trained using XGBoost one to predict the top-level category of a product and another for the bottom-level category. These models learn from the TF-IDF features to make accurate predictions based on patterns in the training data.

When it's time to make predictions on new, unseen data, the test dataset follows the same path: preprocessing, vectorization, and then classification using the trained models. Finally, the predictions for both top and bottom categories are combined and saved into a .parquet file format, which is efficient for storage and easy to use in later stages of analysis or submission.

III. Results

This section presents the performance evaluation of the models developed to classify Etsy product listings into both top-level and bottom-level categories. We report key classification metrics including accuracy, precision, recall, and F1 score. The goal is to assess the effectiveness of different machine learning models and feature representations for handling both coarse-grained and fine-grained categorization tasks.

A. *Top-Level Category Results*

We first evaluated the models on the task of predicting top-level product categories. This task involves fewer labels and is comparatively less affected by class imbalance. The models compared include XGBoost, Random Forest, Logistic Regression, and a baseline Multinomial Naive Bayes classifier, all using TF-IDF features.

As shown in Table I XGBoost demonstrated the strongest performance across all metrics, achieving the highest weighted F1 score. Logistic Regression and Random Forest also showed reasonable results but trailed behind XGBoost. The baseline Naive Bayes model performed significantly lower, validating the need for more advanced classifiers for this task.

TABLE I
Performance Metrics for Top-Level Category Prediction

Model	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
TF-IDF + XGBoost	**83.0**	**84.5**	**82.0**	**82.8**
TF-IDF + Random Forest	72.0	74.1	71.2	72.1
TF-IDF + Logistic Regression	65.0	67.8	63.5	65.0
Multinomial Naive Bayes (Given Baseline)	63.9	65.2	62.8	63.9

B. *Bottom-Level Category Results*

Bottom-level category prediction posed greater challenges due to increased label granularity and class imbalance. The number of distinct categories is significantly higher, and some classes are underrepresented in the dataset.

Among the models evaluated, XGBoost again performed the best, achieving the highest F1 score. Logistic Regression showed moderate performance, while Random Forest struggled considerably on this task, likely due to its limitations in handling high-dimensional sparse features and imbalanced distributions. The performance of the evaluated models for bottom-level category prediction is reported in Table II.

TABLE II
PERFORMANCE METRICS FOR BOTTOM-LEVEL CATEGORY PREDICTION

Model	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
TF-IDF + XGBoost	**58.9**	**62.3**	**57.4**	**58.6**
TF-IDF + Random Forest	30.0	34.1	28.5	30.1
TF-IDF + Logistic Regression	46.0	48.7	44.2	46.0

C. *Comparison of Pre-trained Embedding Approaches*

To understand how our model compares with existing methods, we evaluated two popular approaches from previous studies that use pre-trained embeddings. One method combines Sentence-BERT embeddings with logistic regression, while the other uses GloVe word embeddings with a random forest classifier. As shown in Table III, we compared these models against our TF-IDF + XGBoost approach for top-level category classification. Due to computational constraints, this comparison was only performed for top categories. Among the three, our XGBoost model outperformed both embedding-based methods, suggesting that traditional feature-based models can still offer strong results when paired with effective preprocessing and classification techniques.

TABLE III
PERFORMANCE METRICS FOR TOP-LEVEL CATEGORY PREDICTION

Model	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
TF-IDF + XGBoost	**83.0**	**84.5**	**82.0**	**82.8**
Sentence-BERT + Logistic Regression [16]	74.8	75.2	72.3	73.5
GloVe + Random Forest [17]	66.1	69.5	62.7	64.4

D. *Visualization of Learned Representations*

To gain insights into how well the learned embeddings capture category semantics, we applied t-distributed Stochastic Neighbor Embedding (t-SNE) to project high-dimensional sentence embeddings into two-dimensional space for visualization.

Fig. 4 shows the t-SNE plot of the top-level category clustering. Each point represents a product listing, and colors indicate the corresponding top-level category. The visualization reveals that several category groups, such as *home_and_living*, *accessories*, and *electronics_and_accessories*, form well-separated and coherent clusters. This indicates that the Sentence-BERT embeddings successfully capture meaningful semantic distinctions among major product groups. Some overlap exists, particularly in categories with semantic similarity (e.g., *toys_and_games* and *craft_supplies_and_tools*), which is expected due to shared vocabulary and tag usage.

In contrast, Fig. 5 presents the t-SNE visualization for the 30 most frequent bottom-level categories. Despite the increased granularity and class imbalance, the clusters remain reasonably well-defined. This reflects the embedding model's ability to encode fine-grained distinctions between product types (e.g., *refrigerator_magnets*, *scrub_caps*, and *cake_stands*). However, some overlap is still evident, especially in closely related subcategories within the same top-level group. This suggests that while embeddings provide strong semantic cues, bottom-level classification may benefit from additional feature engineering or hierarchical modeling.

Together, these visualizations validate the effectiveness of sentence embeddings in capturing latent structure in product metadata, offering both interpretability and practical insights for improving classification and recommendation systems.

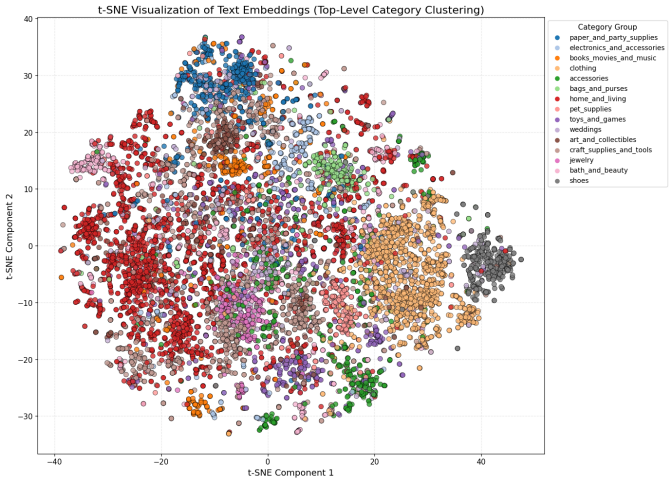

Fig. 4. **t-SNE visualization of text embeddings for the top-level categories.**

IV. CONCLUSION

This study tackled the challenge of classifying Etsy product listings into top- and bottom-level categories using noisy, user-generated metadata. By combining robust preprocessing with TF-IDF vectorization and ensemble models like XGBoost, we achieved promising results. The key findings of the project are:

- **Top-Level Categories:** XGBoost achieved 83% accuracy and a weighted F1 score of 0.828, showcasing its strength with structured textual features.
- **Bottom-Level Categories:** While more challenging due to label granularity and class imbalance, XGBoost still

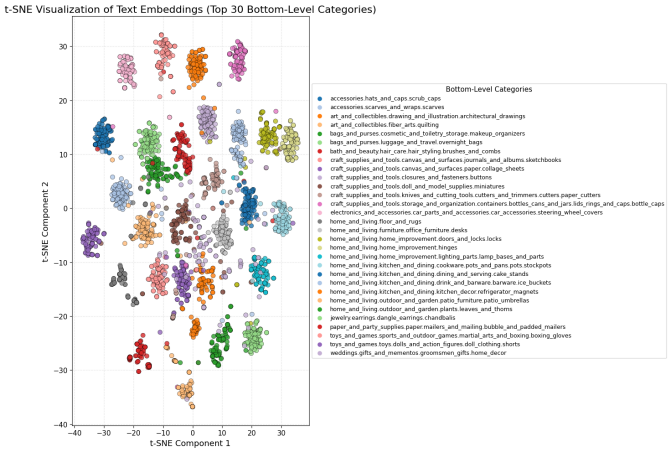

Fig. 5. **t-SNE visualization of text embeddings for the top 30 bottom-level categories.**

outperformed other models, reaching a weighted F1 score of 58.6%.

- **Feature Importance:** Titles and tags were the most predictive features, emphasizing the value of textual metadata in product classification.

These findings highlight the importance of preprocessing, feature engineering, and choosing the right model architecture in building scalable classification systems for e-commerce.

A. *Future Work*

Future work will focus on improving bottom-level category prediction by mitigating class imbalance through SMOTE or cost-sensitive learning. Fine-tuning transformer models like BERT on Etsy-specific data could enhance semantic understanding. Additionally, combining structured metadata with pre-trained embeddings in hybrid architectures, and improving computational efficiency through GPU acceleration or model compression, will support scalable deployment in real-world applications.

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